Introduction

Gene Set Enrichment Analysis (GSEA) tests whether a gene set is concentrated at the top or bottom of a ranked gene list (e.g., by log fold change or signed p-value). Unlike over-representation analysis, it uses the full ranking and is more sensitive to coordinated modest changes within a pathway.

Prerequisites

Ranked gene statistic (LFC or signed -log10 p); a curated gene-set database (MSigDB, Reactome, KEGG).

Theory

GSEA walks down the ranked list, incrementing a running-sum statistic when encountering a set member (weighted by the rank statistic) and decrementing otherwise. The **enrichment score (ES)** is the maximum deviation from zero; a permutation-based null distribution yields p-values. `fgsea` uses a fast adaptive multi-level permutation scheme.

Normalised ES (NES) adjusts for gene-set size and allows cross-set comparison.

Assumptions

The ranking reflects biological signal; gene sets are well-curated; permutation null captures the null distribution of ES.

R Implementation

```r
library(fgsea); library(msigdbr)

set.seed(2026)
n_genes <- 5000
genes <- paste0("g", 1:n_genes)
stats <- rnorm(n_genes); names(stats) <- genes

# Inject a pathway signal
pathway_A <- sample(genes, 30); stats[pathway_A] <- stats[pathway_A] + 1.5
pathway_B <- sample(genes, 20); stats[pathway_B] <- stats[pathway_B] - 1.2

pathways <- list(pathway_A = pathway_A, pathway_B = pathway_B,
                 pathway_C = sample(genes, 40))

fg <- fgsea(pathways = pathways, stats = sort(stats, decreasing = TRUE),
            minSize = 10, maxSize = 500)
fg[, .(pathway, NES, pval, padj)]
plotEnrichment(pathways$pathway_A, sort(stats, decreasing = TRUE))
```

Output & Results

A table with NES, p-value, and adjusted p-value per pathway. The enrichment plot for pathway_A shows a strong positive leading edge.

Interpretation

"GSEA against MSigDB Hallmark revealed positive enrichment for inflammatory response (NES = 1.8, BH-FDR < 0.01) and negative enrichment for oxidative phosphorylation (NES = -1.5)."

Practical Tips

- Use the `stat` column from DESeq2 or `t` from limma as the ranking; signed -log10(p-value) is an alternative.
- Remove NAs from the ranking; fgsea is sensitive to ties and NAs.
- Report the **leading-edge subset** for biologically driven interpretation.
- MSigDB has Hallmark (50 curated), C2 (curated pathways), C5 (GO), C7 (immune); match collection to biological question.
- GSEA tests concentrated enrichment, not direction-specific effects; combine with ORA for complementary evidence.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat